# Mustafa Özen

Mustafa Özen | Google Scholar | ozen.mustafa2025@gmail.com | +12272297393

## Education

**University of Maryland, College Park** — Aug 2026 – Jun 2028
M.S. in Artificial Intelligence — College Park, Maryland

**Sabancı University** — Sep 2022 – Jun 2026
B.Sc. in Computer Science and Engineering — Istanbul, Turkey
50% Merit-Based Scholarship, GPA: 3.63/4.00

## Publications

Kaymaz, S. V., **Özen, M.**, Çelik, S., Tanrıseven, S., Öktem Olgun, E. E., Canlı, O., Güzel, B., Sarıkaya, Y., Kurt, H., & Yüce, M.
*Hexagonal Au Nanostructure SERS Metasurface for AI-Driven Detection of Pesticide Residues in Real Food Samples. ACS Applied Nano Materials*, 2026. DOI: 10.1021/acsanm.6c00433

Kaymaz, S. V., **Özen, M.**, Çelik, S., Tanrıseven, S., Öktem Olgun, E. E., Canlı, O., Güzel, B., Sarıkaya, Y., Kurt, H., & Yüce, M.
*Smart Pesticide Detection via Hexagonal MIM Metasurfaces Integrated with Deep Feed-Forward Networks.*
Oral presentation at *EurasianBioChem 2025*, Ankara, Turkey.

Sağlam, H. C., **Özen, M.**, & Yanıkoğlu, B.
*ADAPT: Adaptive Domain Alignment via Self-Supervised Pre-Training.*
Manuscript in preparation for submission to *CVPR 2027*.

## Research Experience

**Research Assistant, Self-Supervised Learning for Dermatology** — Oct 2025 – Present

- Developing **ADAPT**, a masked self-distillation framework for learning transferable representations from clinical and dermoscopic images under the supervision of Prof. Ayşe Berrin Yanıkoğlu.
- Evaluated classification, linear probing, and segmentation performance against state-of-the-art self-supervised methods, including SimMIM, iBOT, and DINOv3; preparing a research manuscript for submission to *CVPR 2027*.

**Research Fellow, TÜBİTAK 2247-C STAR Program** — Jan 2025 – Mar 2026

- Carried out research on Raman spectroscopy (SERS) for pesticide residue analysis at Sabancı University under the Scientific and Technological Research Council of Türkiye (TÜBİTAK) 2247-C STAR fellowship.
- Developed deep learning pipelines combining spectral feature extraction, data augmentation, compact CNN architectures, and multitask learning for robust pesticide classification and simultaneous quantification.

**Research Project: Semantic Segmentation for Autonomous Driving** Oct 2024 – Feb 2025

- Conducted a comprehensive study of semantic segmentation for autonomous driving on the Cityscapes dataset, experimenting with DeepLabv3+, U-Net, and FCN architectures to compare computational cost and segmentation quality.
- Analyzed performance across varying backbone configurations to evaluate generalization under real-world driving conditions.
- Refined network architectures and inference strategies to balance accuracy with real-time processing efficiency.

**Research Project: Image Classification with Mamba Vision** Jun 2024 – Nov 2024

- Investigated Mamba Vision (state-space) models as lightweight and efficient alternatives to CNNs and Vision Transformers.
- Trained and benchmarked networks in PyTorch, demonstrating favorable accuracy–efficiency trade-offs.
- Performed ablation analyses to evaluate the influence of architecture depth, attention span, and spectral feature interactions.

## Work Experience

**Learning Assistant, Sabancı University** Oct 2023 – Jun 2026

- Supported undergraduate courses including *Computer Vision*, *Data Structures*, *Programming Fundamentals*, and *Calculus II* through labs, office hours, tutorials, grading, and student feedback.
- Designed tutorial materials and video content for *EE-417 Computer Vision*, covering advanced PyTorch concepts, model structuring, and training workflows.
- Assisted with homework development, laboratory sessions, exam administration, and course coordination across multiple teaching appointments.

**Computer Vision Intern, BAYKAR Technologies** Jun 2025 – Aug 2025

- Participated in research on UAV-based autonomous landing systems, emphasizing runway segmentation and pose estimation from onboard imagery.
- Developed semantic segmentation pipelines integrated with classical (PnP + RANSAC) and deep learning–based localization techniques.
- Worked alongside engineering teams to embed perception modules into UAV navigation systems, enhancing landing reliability under GNSS-denied conditions.

## Skills

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| **Programming Languages** | Python, C++, C, SQL, Java |
| **Libraries & Frameworks** | PyTorch, NumPy, OpenCV, Scikit-learn, TensorFlow |
| **Languages** | English (Advanced, TOEFL iBT 102), Turkish (Native) |
| **Tools & Platforms** | Git, Docker, Google Colab, Overleaf |
| **Mathematical Foundations** | Linear Algebra, Numerical Analysis, Differential Equations, Abstract Algebra, Number Theory |